

Euclid's Elements — Book I

Proposition 31

Formal Reconstruction — Technical Documentation

Jurek Rózański
Military University of Technology, Warsaw, Poland
`jerzy.rozanski@wat.edu.pl`

CGJTEAMLAB

Constructing a parallel line through a given point

1 Introduction

Proposition I.31 is a construction theorem: given a straight line and a point outside it, construct a straight line through the point parallel to the given line.

At the level of Euclid's sequence the proposition follows I.29 and I.30, but its logical status is different. The construction itself does not require a Euclidean uniqueness principle for parallels. It transfers a direction by copying an angle and then invokes the neutral alternate-angle criterion of I.27.

In the present Lean development the Book I theorem is correspondingly small. The substantial geometry is already isolated in the reusable interface theorem `hilbert_parallel_through_point_exists`. Proposition I.31 is a thin wrapper around that result.

2 Classical Configuration

Euclid's literal configuration uses different labels from the project-level statement. Let A be the given external point and let BC be the given straight line. Choose a point D on BC and join AD . At A construct, by I.23, an angle equal to $\angle ADC$, and extend the constructed ray to a straight line EAF .

Thus the classical dependency is

$$\text{I.23} \longrightarrow \text{equal alternate angles} \xrightarrow{\text{I.27}} EAF \parallel BC.$$

The construction uses no right angle. The copied angle may have any nondegenerate size; its numerical measure is irrelevant.

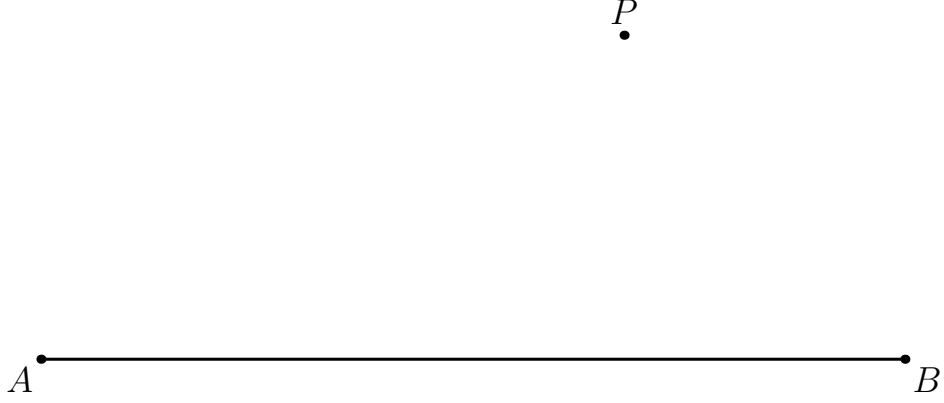


Figure 1: The classical configuration of Proposition I.31. The point D is chosen on the given line BC , the transversal AD is joined, and the line EAF through the external point A is constructed so that $\angle EAD \cong \angle ADC$. Proposition I.27 then identifies EAF as parallel to BC .

3 Why No Right Angle Is Needed

A familiar modern construction of a parallel can proceed through two perpendiculars, but this is not Euclid's proof of I.31. The relevant mathematical mechanism is simply direction transfer by angle congruence.

The point is also important for the logical organization of the Hilbert reconstruction. Existence of a parallel through an external point can be obtained from angle construction and the neutral converse alternate-angle theorem. Uniqueness of that parallel is additional Euclidean content.

Consequently the natural schematic distinction is

$$\boxed{\text{parallel existence in neutral geometry}} \\ \text{versus} \\ \boxed{\text{parallel uniqueness in the Euclidean layer}}.$$

4 The Hidden Formal Configuration

The project-level theorem uses the labels A, B, P for the given line and the external point. Its statement asks for a point Q such that

$$P \neq Q, \quad AB \parallel PQ.$$

The reusable proof contains more order and side information than this final statement displays.

First, the line AP is used as the transversal. An interior point E is chosen with

$$A - E - P.$$

The proof then introduces a second auxiliary point S by extending BA beyond A :

$$B - A - S.$$

Because A lies on the transversal AP , while B and S lie on opposite sides of it, the proof obtains

$$\text{OppositeSide}(B, S; AP).$$

This point S is a genuine hidden witness in the current Lean proof. It is not present in the public theorem statement, but it is the geometric device that selects the correct side on which the copied angle is constructed.

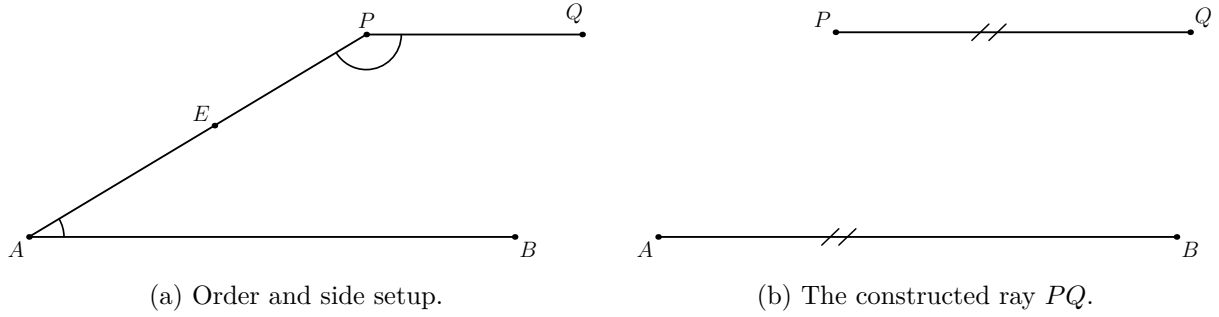


Figure 2: The formal geometry hidden behind the thin Book I wrapper. In panel (a), $A - E - P$ fixes the transversal rays and $B - A - S$ places the auxiliary point S on the side of AP opposite to B . In panel (b), angle construction produces Q on the same side of AP as S with $\angle EAB \cong \angle EPQ$. Hence B and Q are on opposite sides of AP , which is exactly the orientation needed by the alternate-angle parallel criterion.

5 Hilbert Reconstruction

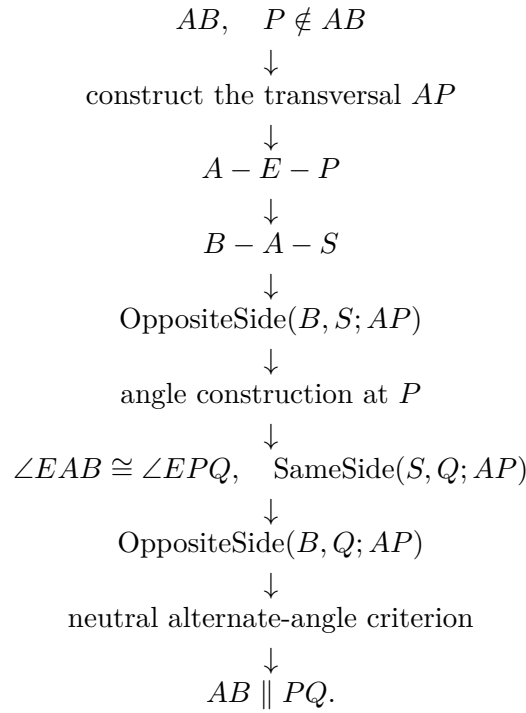
The substantial theorem is

```

theorem hilbert_parallel_through_point_exists
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
  (hAB : A != B)
  (hABP : not Collinear Geo A B P) :
  exists Q : Geo.Point,
    P != Q /\
    Geo.Parallel A B P Q

```

Its current proof has the following geometric architecture:



The side transport is therefore not decorative bookkeeping. It converts the side selected for the temporary witness S into the orientation data for the actual constructed point Q .

There is no geometric case split in this proof. The construction proceeds through a single sequence of witnesses and side relations.

6 Lean Formalization

The current production file is

```

import CGJteamLab.HilbertInterface

namespace Geometry

universe u

variable {Geo : Geometry.Geo.{u}}

theorem euclid_proposition_31
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
  (hAB : A != B)
  (hABP : not Collinear Geo A B P) :
  exists Q : Geo.Point,
    P != Q /\
    Geo.Parallel A B P Q := by

exact
  hilbert_parallel_through_point_exists
    Geo

```

```

      A B P
      hAB
      hABP
end Geometry

```

Thus `euclid_proposition_31` is a genuine thin wrapper. It contains no proposition-local helper theorem and introduces no additional geometric witness beyond those already produced by the interface theorem.

The direct dependency of the Book I theorem is therefore

```
euclid_proposition_31  →  hilbert_parallel_through_point_exists.
```

The deeper construction uses incidence for the transversal, order for E and S , Hilbert angle construction for Q , side-of-line transport, and the neutral alternate-angle criterion.

The public typeclasses are

```

[HilbertIncidence Geo]
[HilbertCongruence Geo]

```

and no `HilbertEuclideanPlane` instance occurs in the proposition statement.

The current source of `Proposition31.lean` contains no `sorry` and no proposition-local `axiom`. No fresh `lake build` or `#print axioms` audit was run as part of this diagrammatic review, so no stronger compilation or axiomatic-status claim is made here.

7 Neutral Existence Versus Euclidean Uniqueness

Proposition I.31 proves existence:

$$\exists PQ \text{ through } P, \quad PQ \parallel AB.$$

It does not prove uniqueness; that belongs to the Euclidean parallel layer. This also clarifies I.29 structurally: an I.31-type construction produces a candidate parallel neutrally, while a uniqueness principle can identify it with an already given parallel. This is not an extra step in Euclid’s proof of I.31.

8 Diagrammatic Audit

The preserved Book1R chapter used three separate figures: initial data, the copied-angle construction, and the final parallel pair. In the revised audit the first and third do not count as new geometric states: the initial picture adds no construction, while I.27 changes only the logical status of the already constructed line PQ .

The revised figure set instead records the two genuinely different layers:

1. the literal classical Euclid/Heath construction;

2. the formal Hilbert construction, split into the order/side state and the subsequent angle-construction state.

The hidden witness S is included because it is a real auxiliary point and encodes the side choice used by the current Lean proof. It was absent from the preserved diagrams even though the text discussed the side condition.

Accordingly the final count is two figures with three panels. No extra panel is used for $AB \parallel PQ$, because that conclusion introduces no new geometry.

9 Conclusion

Proposition I.31 transfers direction by constructing an equal alternate angle and invoking I.27. The Hilbert reconstruction exposes the order and side data suppressed by the classical picture, most notably the hidden witness S that selects the correct side of the transversal. The Book I theorem `euclid_proposition_31` is a thin wrapper around the reusable neutral existence theorem `hilbert_parallel_through_point_exists`; uniqueness of the parallel belongs to the later Euclidean structure.